

RS004 Week 7 — English Worksheet

Topic: Platform Collision · **Course:** Platformer Game · **Time:** about 45 minutes

This week you think about how a player can **land on platforms**. You read code that keeps platforms in a **list**, checks each one with `colliderect()`, and only lands when the player is falling *down* onto the top of a platform. Then you explain the collision code from today's lesson in your own words.

Keep these words handy: **platform**, **list**, `colliderect()`, **land**, **falling** ($vy > 0$).

1 · Predict 🧠

Read each snippet. Write what you think will happen.

```
platforms = [
    pygame.Rect(0, 550, 800, 50),
    pygame.Rect(200, 450, 120, 20),
]
for plat in platforms:
    pygame.draw.rect(screen, BROWN, plat)
```

How many platforms get drawn? What does the `for` loop do here?


```
if player.colliderect(plat) and velocity_y > 0:
    player.bottom = plat.top
    velocity_y = 0
    on_ground = True
```

Why check `velocity_y > 0`? What does `player.bottom = plat.top` do?


```
on_ground = False
for plat in platforms:
    if player.colliderect(plat) and velocity_y > 0:
        ...
```

Why is `on_ground` set to `False` *before* the loop, every frame?

2 · Why `velocity_y > 0`? 🤔

The player can touch a platform while moving up *or* down. We only want to land when moving **down**.

What would go wrong if we removed the `velocity_y > 0` check and the player jumped up through a platform?

3 · Spot the Bug 🐛

Each snippet is broken. Read it, write the fixed code on paper, then explain why the original was wrong.

Bug A — The player should land on every platform. Right now it only ever lands on the first one.

```
plat = platforms[0]
if player.colliderect(plat) and velocity_y > 0:
    player.bottom = plat.top
    velocity_y = 0
    on_ground = True
```

Hint: we check only one platform. How do we check *all* of them?

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — The player should stand on top of a platform. Right now it sinks halfway into it.

```
for plat in platforms:
    if player.colliderect(plat) and velocity_y > 0:
        velocity_y = 0
        on_ground = True
```

Hint: nothing snaps the player's bottom to the platform top.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — `on_ground` should be `True` only when actually standing on something. Right now it gets stuck `True` even after the player walks off a platform into the air.

```
on_ground = True
for plat in platforms:
    if player.colliderect(plat) and velocity_y > 0:
        player.bottom = plat.top
        velocity_y = 0
```

Hint: start each frame assuming the player is *not* on the ground.

Write the fixed code:

Why was it wrong? Why does your fix work?

4 · Explain the Code

Read this working collision code carefully, then answer the questions about it.

```
platforms = [  
    pygame.Rect(0, 550, 800, 50),  
    pygame.Rect(200, 450, 120, 20),  
]  
  
on_ground = False  
for plat in platforms:  
    if player.colliderect(plat) and velocity_y > 0:  
        player.bottom = plat.top  
        velocity_y = 0  
        on_ground = True
```

Why are the platforms kept in a list instead of separate variables?

What does `for plat in platforms:` do on each pass through the loop?

What does `player.colliderect(plat)` return, and how is it used in the `if`?

What does the line `player.bottom = plat.top` do to the player's position?

Why is `on_ground = False` written before the loop instead of inside it?

5 · Explain Your Lesson Code 🎥

Today in your live lesson you wrote collision code for your player. Record a short phone video explaining the code **you** wrote. You may show it running. Try to use these words: **platform, list, colliderect, land, falling**.

1. Point to where you keep your platforms in a list.
2. Read your collision check out loud and explain `velocity_y > 0`.
3. Show the player landing on a platform, then walking off and falling.
4. Explain one part of the code that was tricky for you.

Write what you will say in your video. Plan it here before you record.

Submit 

Send this worksheet + a video explaining your lesson code to teacher on KakaoTalk.